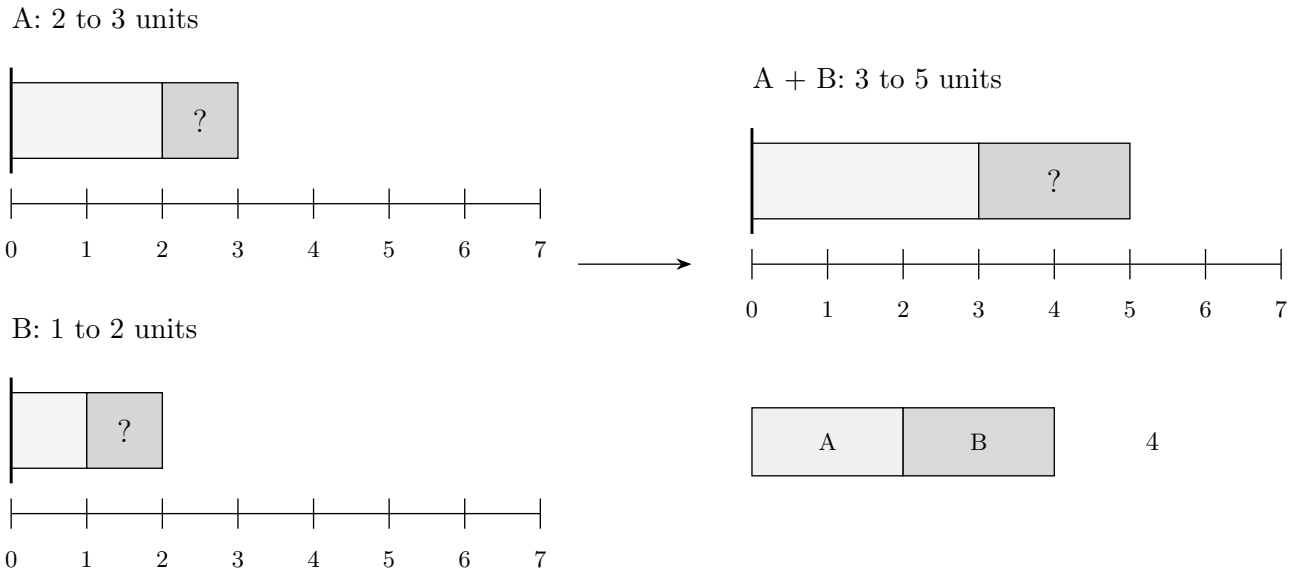
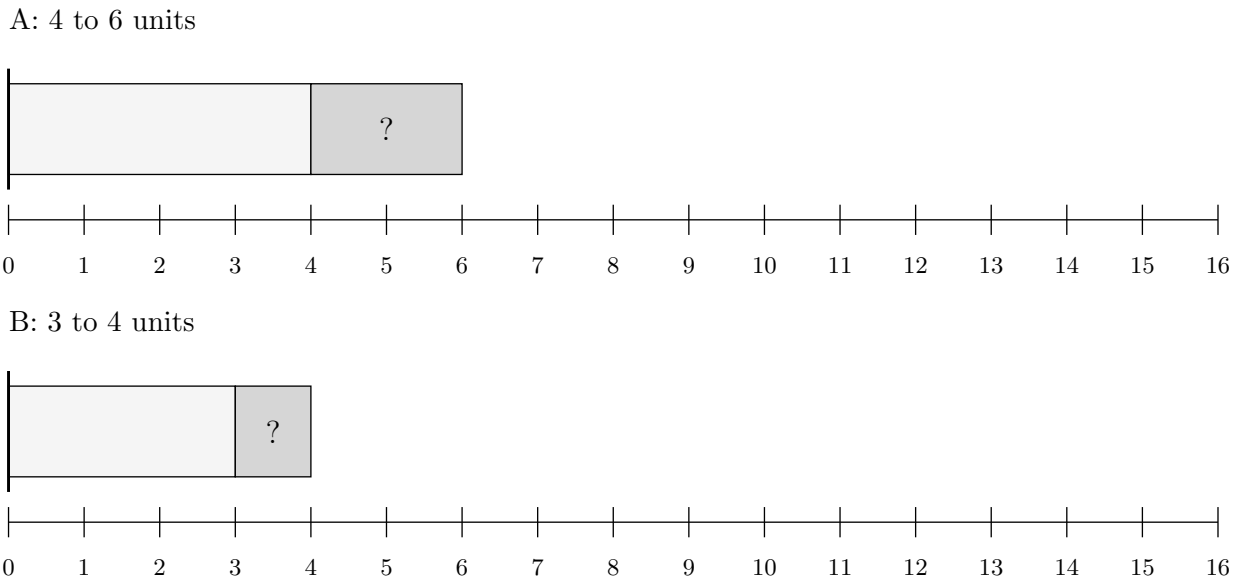
A hidden end can be anywhere in its gray band, including the edges. Each trial may use new lengths. Join strips end to end from 0, with no gap or overlap.

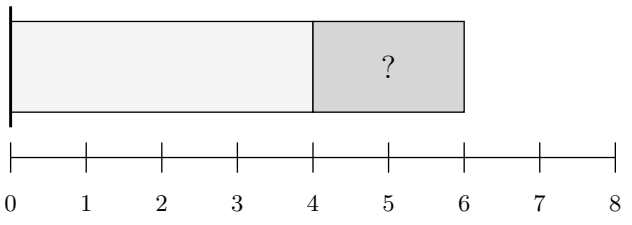


Problem 1: Find the tightest guaranteed range for the join. Someone claims it must end between 8 and 9; build allowed strips that defeat that claim.

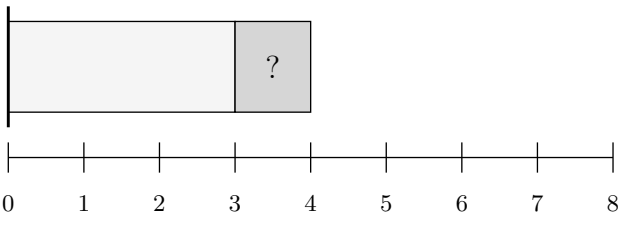


Problem 2: Try only A with B, C with D, and E with F. Choose a pair whose join always reaches 8 and fits within 10. For each rejected pair, let a partner defeat one promise with allowed lengths.

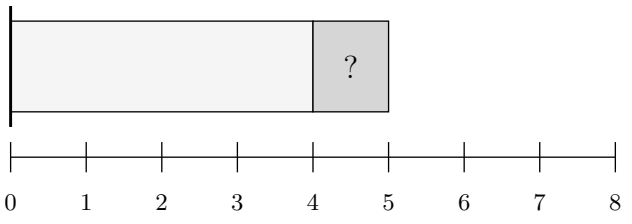
A: 4 to 6 units



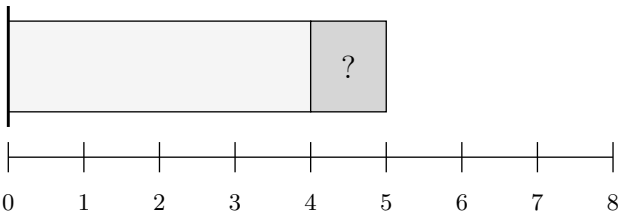
B: 3 to 4 units



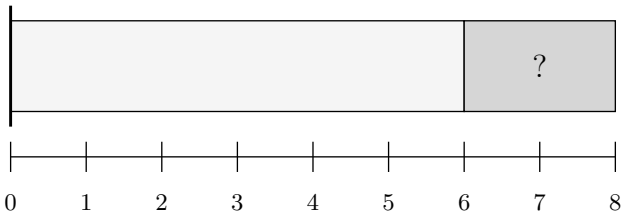
C: 4 to 5 units



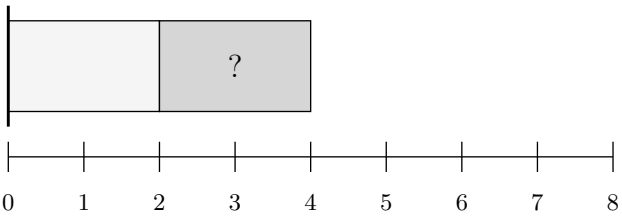
D: 4 to 5 units



E: 6 to 8 units



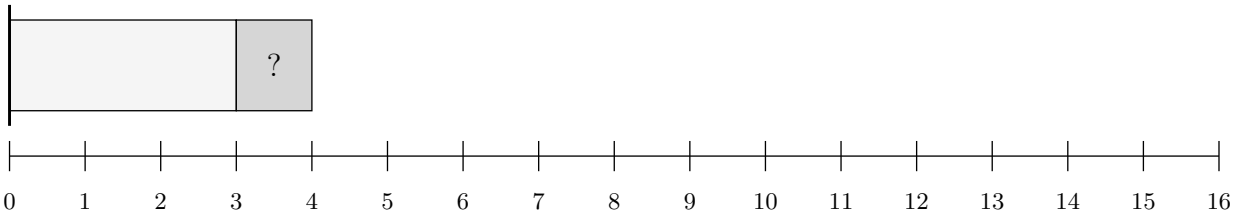
F: 2 to 4 units



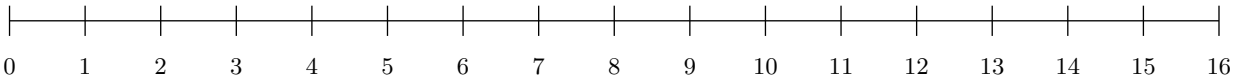
Reaching 8 includes ending at 8. Fitting within 10 includes ending at 10.

Problem 3: A is 3 to 4 units long. Make a range card for B so every join reaches 7 and fits within 9. Make B's allowed range as wide as possible, and explain why it cannot be wider.

A: 3 to 4 units



B:





Problem 4: A different pair has A from 4 to 6 and B from 3 to 4. You may learn one exact length. Which strip would you uncover to leave the narrowest guaranteed range for the join?